Mohaimen Khan

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EXPERIENCE

Senior Software Engineer | Microsoft (GitHub) - Applied Science and Model Evaluation at GitHub Copilot

January 2026 - Present

- Engineered cross-platform grading and run-level orchestration across Linux, Windows, and self-hosted-compute environments for an LLM evaluation platform supporting **300K+ submitted runs and 10M+ task evaluations per month**.
- Implemented LLM-as-judge privacy evaluation and integrated security and benchmark-cheating graders into automated evaluation workflows, producing **10M+ assessed pass/fail judgments per month**.
- Designed and implemented a self-service grader registry using SQL-backed metadata, role-based access control, and immutable artifacts; added CLI publishing and worker-side package verification for reproducible, version-pinned evaluations.
- Developed a Harbor-native benchmark using **58 real-world Windows tasks**, preserving task instructions, environments, reference solutions, and verifiers; Currently generates evals across VSCode Copilot, Copilot CLI, Copilot App and other products across the company.
- Productionized agentic benchmark analysis through run-level graders and scheduled evaluations, generating **2,600+ successful full-run reports per month** with structured task-outcome, cost, and execution-time metrics.
- Delivered service-authoritative run and queue observability across backend APIs, CLI, and web dashboards, separating queue times, execution, and completion timing to help engineers diagnose stalled evaluations.
- Corrected multi-agent token accounting across telemetry ingestion, CLI reports, and model-comparison dashboards by parsing nested trajectories and preserving session identity, incorporating previously omitted subagent usage into efficiency comparisons.
- Built reproducible release pipelines for Python evaluation-agent graders across **different platforms/provider variants** to support Windows/Linux execution without runtime package-index access.

Senior Software Engineer | Microsoft - GitHub Copilot for Azure Compute (W365 / Dev Center)

April 2024 - December 2025, Redmond, WA

- Designed and built an LLM-powered orchestration service using Azure OpenAI and Cosmos DB, translating natural-language requests into validated, user-approved execution plans for cloud workstations; delivered the initial VS Code and Visual Studio experience with a **two-engineer team in one quarter**.
- Co-developed Model Context Protocol (MCP) tools for cloud-workstation discovery and management, then added task planning and approval capabilities to enable AI agents to initiate and monitor remote workflows.
- Developed a VS Code extension integrating Azure AI Search, LLM-assisted query rewriting, and Copilot Chat, giving engineers natural-language access to code, documentation, commits, and pull requests.
- Built the cloud-workstation integration for a cross-platform Electron app, using secure tunnels and WebSocket streaming to surface workspace events and Copilot CLI output across Windows, macOS, and Linux.
- Prototyped agent-driven workflows for repository setup, builds, tests, and .NET/Java upgrades using remote execution and secure tunnels, demonstrating long-running coding-agent automation.
- Designed and deployed a feature-flagged Cosmos DB change-feed service to consolidate project metadata across Azure regions, creating a foundation for faster workstation discovery.
- Migrated a Dev Box Azure Portal extension to SDKv2 and restored localization through OneLoc, modernizing the portal integration and addressing recurring on-call issues.
- Co-developed reusable AI-assisted workflows for infrastructure buildouts and versioning, standardizing operational tasks in developer IDEs; the project won a **Microsoft Hackathon Popular Choice award**.

Software Engineer - II | Microsoft - GitHub Copilot for Visual Studio

August 2022 - April 2024, Redmond, WA

- Co-developed a semantic-context prototype using indexed code symbols and token-efficient code skeletons; in a **91-case offline comparison, 44.0%** of responses met both “no hallucination” and “problem solved” criteria, versus **31.9%** without the prototype.
- Built in-process and out-of-process Visual Studio indexing services and integrated indexing with solution-load events, enabling Copilot to access repository-specific code context.
- Expanded offline evaluation of semantic-context quality to **500+ cases across four repositories** and developed custom assertions to detect hallucinations before customer testing.
- Exposed **14 previously masked model regressions** by isolating test context after discovering that generated code could modify tests and conceal failures.
- Implemented deduplication of overlapping document context and prototyped query-aware symbol ranking using Jaccard similarity and embeddings, prioritizing relevant code within limited LLM context windows.

- Built an Azure Blob Storage library using perfect hashing and bulk writes for high-volume code-reference and hover data; secured storage access using Azure managed identities and RBAC.
- Cut snapshot-lookup network calls from **four to one per request—a 75% reduction**—using Git commit IDs, enabling navigation on new branches without waiting for branch-specific indexing.
- Prevented redundant code-index ingestion by checking prior document state, reducing reported queue volume from approximately **100,000 to 4,000 messages per day**.

Software Engineer | Amazon - Amazon Learning Products

October 2021 - July 2022, Seattle, WA

- Developed and implemented the infrastructure for API Gateway, leveraging AWS-CDK resources, allowing 3 million Amazon-Learn users to onboard and access downstream microservices for a SaaS Learning Management Systems product.
- Implemented authentication and lead efforts to implement custom authorization models on top of the API Gateway service using TypeScript, AWS Cognito, and IAM roles allowing development of secured authentication, multi-tenancy, and role-based authorization.
- Developed and deployed a frontend application, using React, TS and AWS pipelines, to optimize costs of a 3rd party learning software which saves Amazon over $4 million annually, a 30% savings compared to the previous model.

Software Engineer | Microsoft - Windows Packaging Service

August 2020 - October 2021, Redmond, WA

- Implemented and extended features on the Windows Packaging and Servicing API, leveraging Azure IaaS and Service Fabric to deliver a geo-redundant, highly reliable update pipeline that packaged, validated, and shipped Windows updates to 1.5 million+ clients every two weeks.
- Designed and implemented an internal .NET service to automatically rotate and update certificates on legacy on-premises servers, reducing service downtime from 2 hours to less than 30 seconds and significantly improving system availability.
- Facilitated a seamless transition to next-generation cloud services by implementing interfaces for on-premises servers containing Windows updating packages while achieving 0% downtime and ensuring uninterrupted service for customers.

SKILLS

Languages: C#, C++, Python, TypeScript, JavaScript, Java, SQL, KQL
AI & Agent Engineering: Azure OpenAI, Azure AI Foundry, agent orchestration, Model Context Protocol (MCP), tool calling, prompt and context engineering, LLM evaluation, graders, benchmark design
Retrieval & Code Intelligence: Azure AI Search, RAG, embeddings, semantic retrieval, code indexing, symbol retrieval
Cloud & Data: Azure, AWS, Cosmos DB, Azure Blob Storage, Azure Data Explorer (Kusto), DynamoDB, PostgreSQL, MongoDB
Frameworks & Developer Tools: .NET, Node.js, React, Electron, VS Code and Visual Studio extensions, Dev Box, Dev Tunnels
Systems & Delivery: Distributed systems, workflow orchestration, microservices, event-driven processing, API design, WebSocket streaming, CI/CD, GitHub Actions, Azure DevOps, Service Fabric

EDUCATION

Bachelor's of Applied Science and Engineering - Computer Engineering | University of Toronto

Minor in Engineering Business, Toronto, ON, Canada

- IEEE Canadian Foundation Scholar, 10k Jeffrey Skoll Scholarship, Vern Gomes Memorial Scholar

PUBLICATIONS

Ai-based code generation using a dynamically constructed symbol graph and code skeletons

Patent, 2025

- Co-inventor of European patent application EP 4647896 A1, assigned to Microsoft Technology Licensing LLC. Describes using dynamically constructed symbol graphs and code skeletons to provide codebase-specific context for AI-assisted code generation. Published November 12, 2025. Patent pending. https://patents.google.com/patent/EP4647896A1/en

System and method for electronic rental platform

Patent, 2021

- Co-inventor of Canadian patent application CA 3090161 A1, assigned to Royal Bank of Canada. Describes an electronic platform that matches renters with properties and estimates suggested bid prices using historical rental data. Published February 14, 2021. Patent pending. https://patents.google.com/patent/CA3090161A1/en